

Muhammad Abdullah Goher

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Education

University of Pennsylvania

Aug 2023 – May 2027

B.S.E. & M.S.E. in Computer Science (Submatriculation), Concentration in Artificial Intelligence

Philadelphia, PA

GPA: 3.72 (B.S.E.) / 3.88 (M.S.E.) | **Coursework:** Operating Systems, Computer Architecture, Databases, Software Design, Algorithms, Data Structures, Interactive Computer Graphics, Big Data Analytics

Experience

AeternalLabs, PBC – Founding Software Engineer

May 2026 – Present

Runtime AI-governance platform (first engineering hire)

Remote

- Built the backend integration layer (FastAPI + Express gateway) connecting a Python scoring engine to live inference, with authentication, rate limiting, and multi-tenant isolation.
- Designed a PostgreSQL persistence layer (SQLAlchemy) with SHA-256 input-integrity hashing and atomic multi-row writes for a tamper-evident audit trail.
- Refactored a ~**6,700-line** React monolith into a modular component architecture and removed provider API keys from the client (top red-team finding).
- Wrote a pytest suite (8 modules, ~1,270 LOC) and parallelized inference calls with a thread pool for a **1.88×** speedup.

Computational Social Science Lab (Prof. Duncan Watts) – Lead Software Engineer Intern

May 2025 – Dec 2025

Nomad – full-stack human-mobility research platform

Philadelphia, PA

- Designed and built Nomad with a custom Express/MongoDB MVC backend exposing secure REST APIs; stored research documents on AWS S3 for scalability.
- Implemented Cypress (frontend) and Vitest (backend) end-to-end tests, parallelizing execution to cut test runtime **50%**.
- Built an automated LLM pipeline extracting structured JSON from **100+** scholarly papers.

Children's Hospital of Philadelphia – Full-Stack Web Developer

Nov 2024 – Aug 2025

Compsy – research platform for behavioral & clinical experiments

Philadelphia, PA

- Built a responsive dashboard and **15+** CRUD data-entry forms in React (Vite) + Tailwind, integrated via REST APIs to a MongoDB backend.
- Implemented stateless JWT authentication (HTTP-only cookies) and context-based data fetching, with responsive behavior across desktop, tablet, and mobile.

Projects

PennOS – Unix-like Operating System | C, POSIX

- Led development of a Unix-like OS with a modular FAT-based file system supporting custom block sizes, mount/unmount, and full file and directory operations.
- Built the userland shell and a kernel with a system-call interface managing process creation, scheduling, and termination alongside file and I/O operations.

Mini Minecraft – Procedural Voxel Engine | C++, OpenGL, GLSL

- Implemented multithreaded chunk loading and terrain generation off the main thread, keeping the world streaming without stalling rendering, plus player physics (collision detection, gravity, jumping, and swimming).
- Built procedural terrain – flora placement via Perlin noise and SDF-based trees – with a dynamic day/night cycle and distance-based fog in GLSL shaders.

West Coast EV Explorer | PostgreSQL, PostGIS, Node.js, Express, React

- Designed a normalized relational schema and materialized views over a **100k+** row database to support fast queries.
- Built backend APIs for zip-code EV lookups, PostGIS nearest-charger search, and state-level comparisons.

Technical Skills

Languages: Python, C, C++, Java, JavaScript/TypeScript, SQL, HTML/CSS

Backend: Node.js, Express, FastAPI, REST APIs, PostgreSQL, MongoDB, SQLAlchemy, JWT auth

Frontend: React, Vite, Tailwind, Redux

Systems, Infra & Testing: C/POSIX, OpenGL, Docker, AWS (S3, EMR), Git, Linux/Shell, Cypress, Vitest, pytest